Manifolds



WEEKEND 4, FRIDAY MORNING

Carlos Cotrini

Friday 23 October 2026

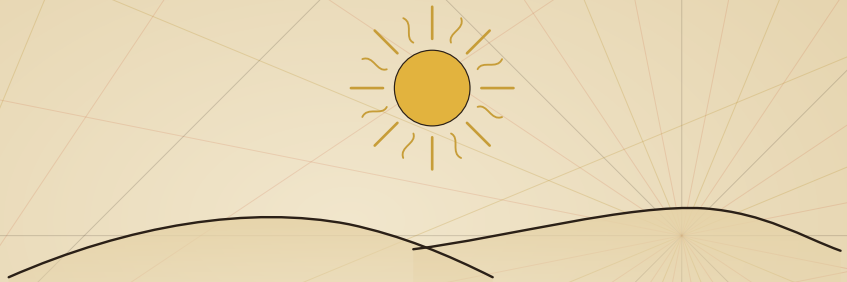


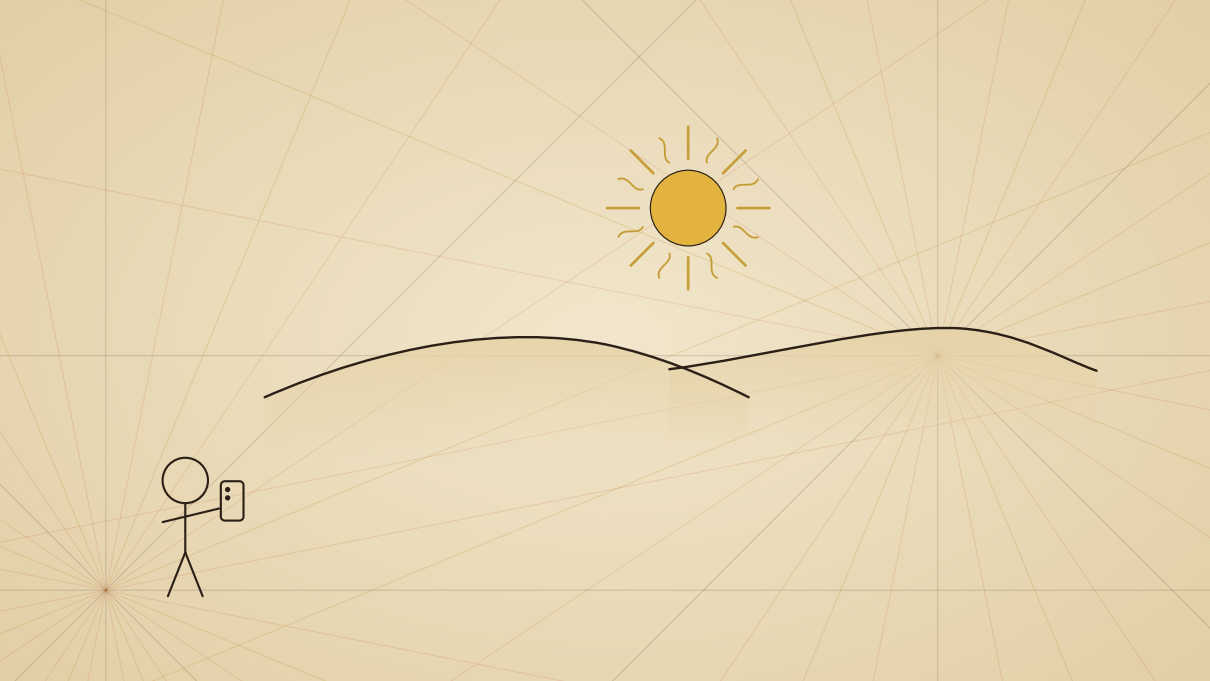


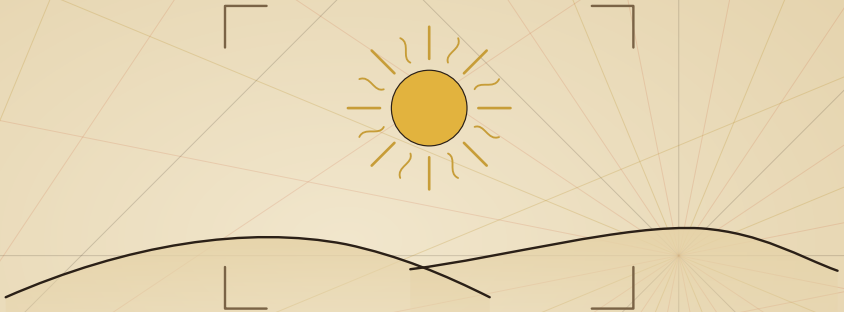




Every morning, the keeper watches the sun rise out of the sea.



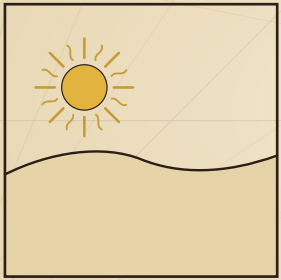




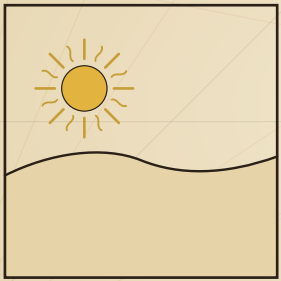


Inland, someone holds up a phone and keeps the moment: a sunrise, saved as millions of pixels.

✳ *Predicting from a photo*



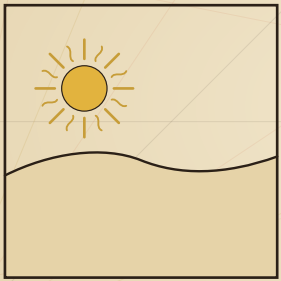
✳ *Predicting from a photo*



REGRESSION

Time of the day

✳ *Predicting from a photo*



REGRESSION

Time of the day

CLASSIFICATION

Season



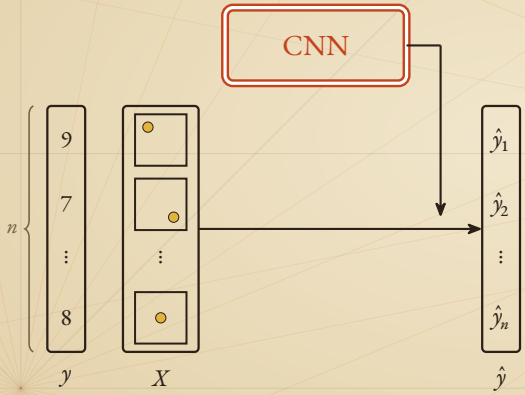
SECTION I

From pixels to coordinates

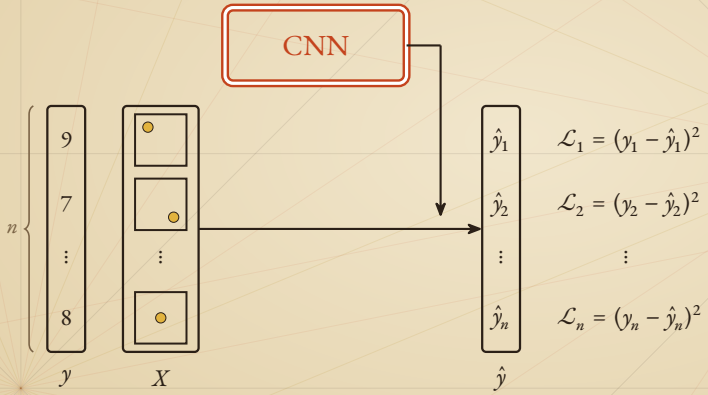
Vapnik's statistical learning



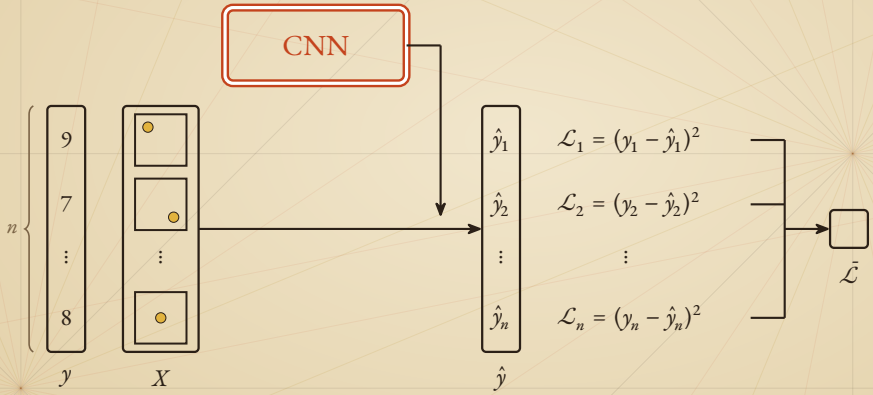
✳ *Vapnik's statistical learning*



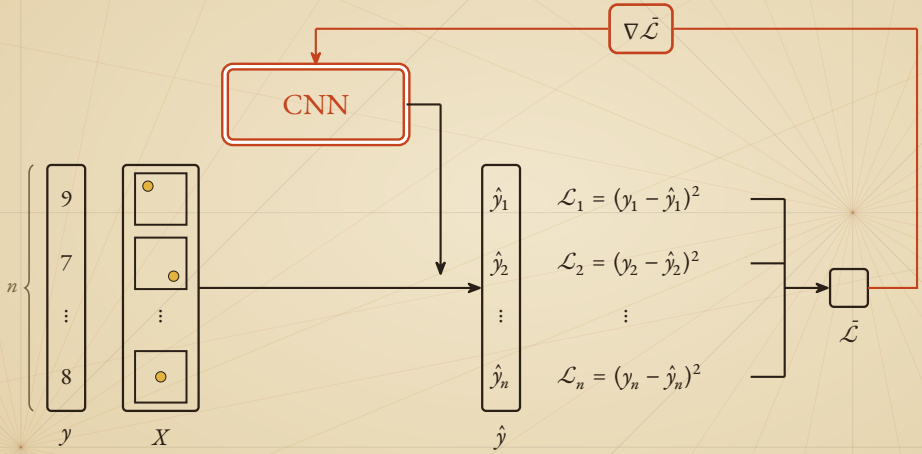
✳ *Vapnik's statistical learning*



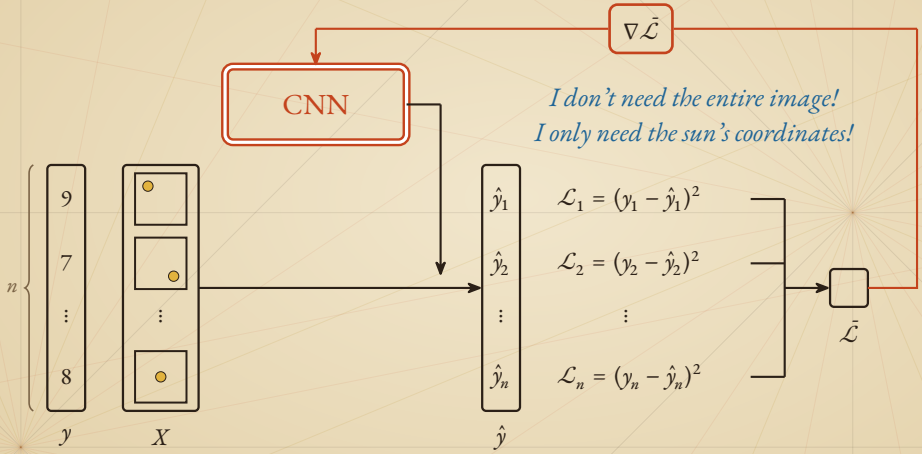
✳ *Vapnik's statistical learning*



✳ *Vapnik's statistical learning*



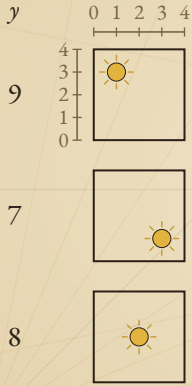
✳ *Vapnik's statistical learning*



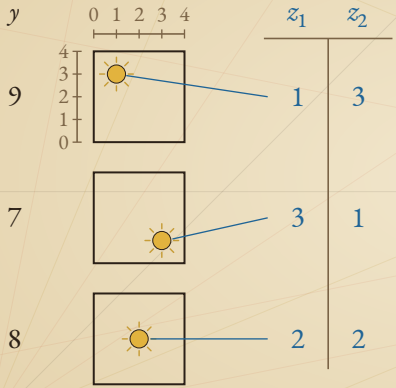
The sun's coordinates



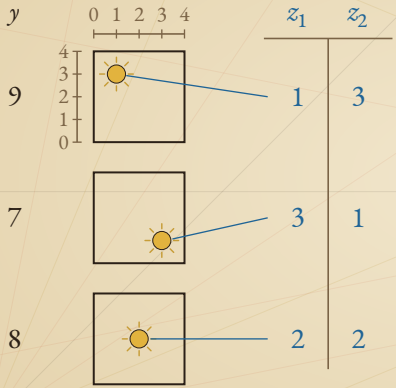
The sun's coordinates



The sun's coordinates

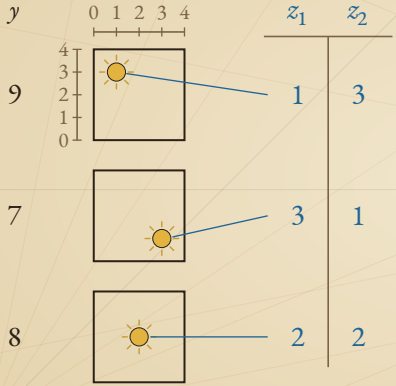


✳ *The sun's coordinates*



$$\hat{y} = w\, z_2 + b$$

✳ *The sun's coordinates*



$$\hat{y} = w\, z_2 + b$$

$$\hat{y}_1 = w \cdot 3 + b$$

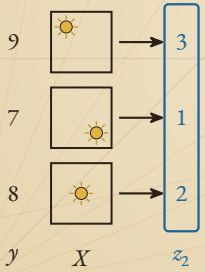
$$\hat{y}_2 = w \cdot 1 + b$$

$$\hat{y}_3 = w \cdot 2 + b$$

Learning w and b

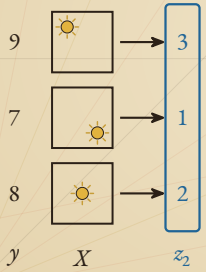


Learning w and b

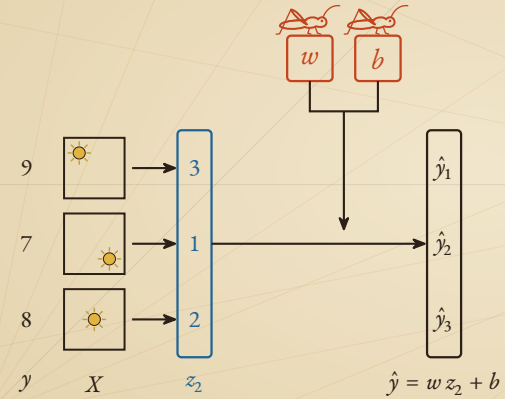


Learning w and b

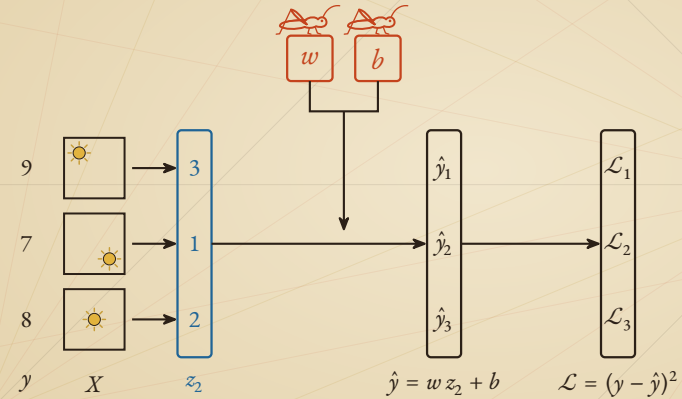




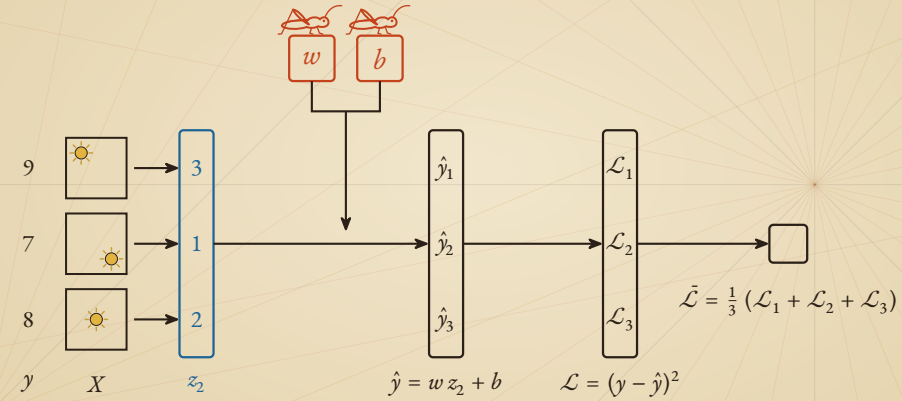
Learning w and b



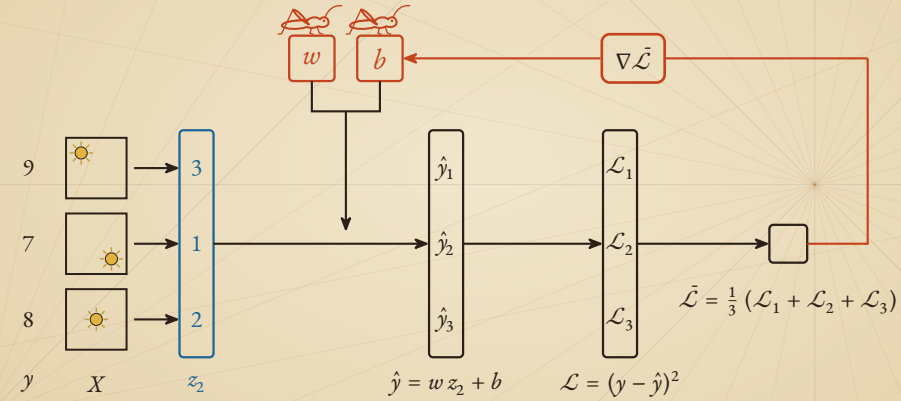
Learning w and b



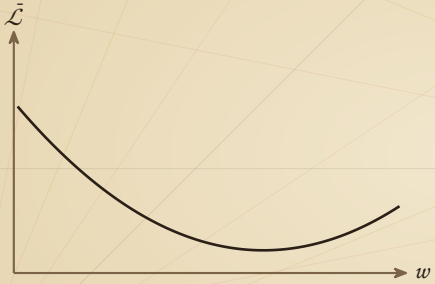
Learning w and b

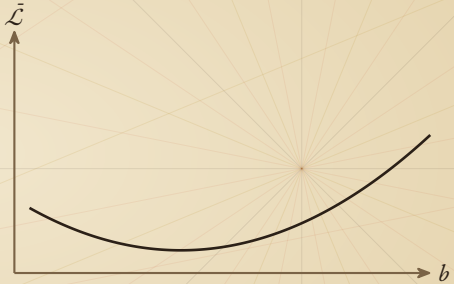


✳ *Learning w and b*

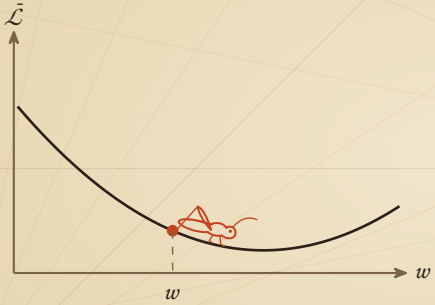


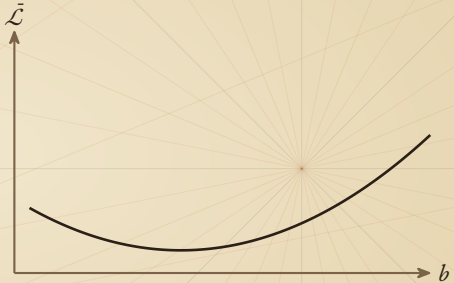
- *Each grasshopper measures its slope*



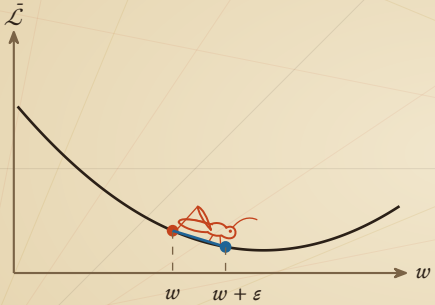


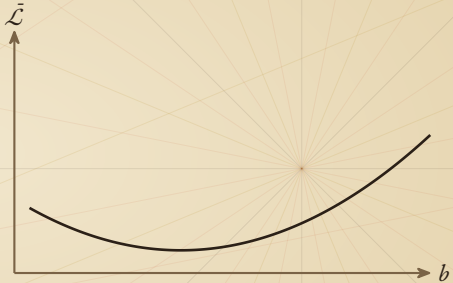
- *Each grasshopper measures its slope*



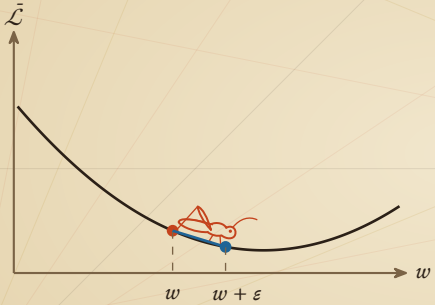


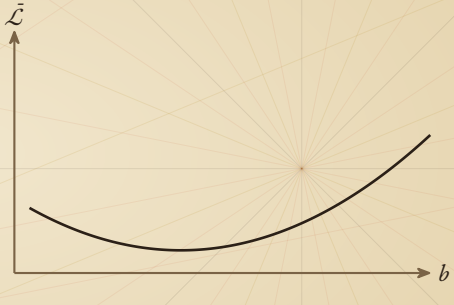
- *Each grasshopper measures its slope*





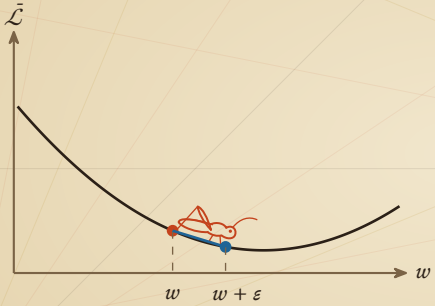
✳ *Each grasshopper measures its slope*

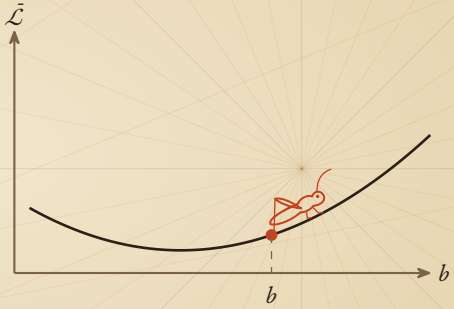




$$\frac{\partial \bar{\mathcal{L}}}{\partial w} \approx \frac{\bar{\mathcal{L}}(w + \varepsilon,\, b) - \bar{\mathcal{L}}(w,\, b)}{\varepsilon}$$

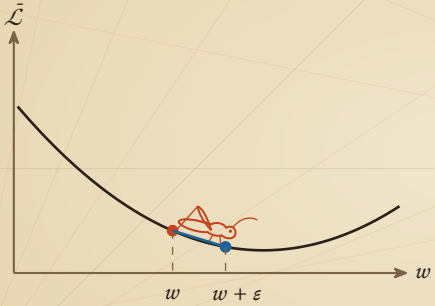
✳ *Each grasshopper measures its slope*

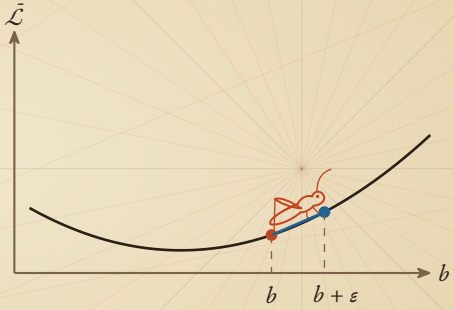




$$\frac{\partial \bar{\mathcal{L}}}{\partial w} \approx \frac{\bar{\mathcal{L}}(w + \varepsilon, b) - \bar{\mathcal{L}}(w, b)}{\varepsilon}$$

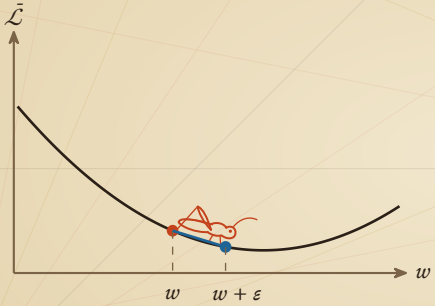
✳ *Each grasshopper measures its slope*



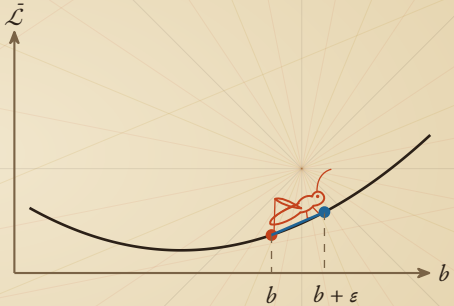


$$\frac{\partial \bar{\mathcal{L}}}{\partial w} \approx \frac{\bar{\mathcal{L}}(w + \varepsilon,\, b) - \bar{\mathcal{L}}(w,\, b)}{\varepsilon}$$

✳ *Each grasshopper measures its slope*



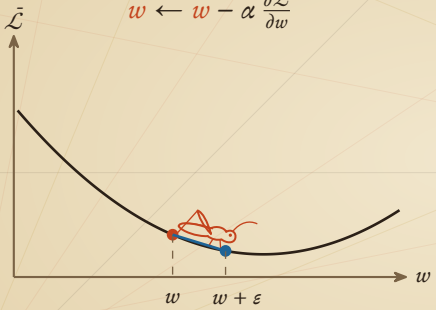
$$\frac{\partial \bar{\mathcal{L}}}{\partial w} \approx \frac{\bar{\mathcal{L}}(w + \varepsilon, b) - \bar{\mathcal{L}}(w, b)}{\varepsilon}$$



$$\frac{\partial \bar{\mathcal{L}}}{\partial b} \approx \frac{\bar{\mathcal{L}}(w, b + \varepsilon) - \bar{\mathcal{L}}(w, b)}{\varepsilon}$$

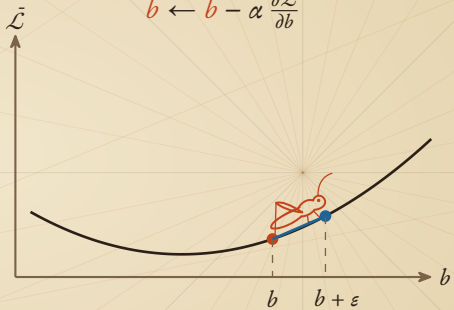
✳ *Each grasshopper measures its slope*

$$w \leftarrow w - \alpha \frac{\partial \bar{\mathcal{L}}}{\partial w}$$



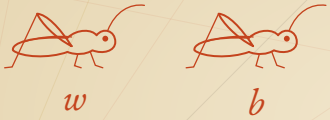
$$\frac{\partial \bar{\mathcal{L}}}{\partial w} \approx \frac{\bar{\mathcal{L}}(w + \varepsilon,\, b) - \bar{\mathcal{L}}(w,\, b)}{\varepsilon}$$

$$b \leftarrow b - \alpha \frac{\partial \bar{\mathcal{L}}}{\partial b}$$



$$\frac{\partial \bar{\mathcal{L}}}{\partial b} \approx \frac{\bar{\mathcal{L}}(w,\, b + \varepsilon) - \bar{\mathcal{L}}(w,\, b)}{\varepsilon}$$

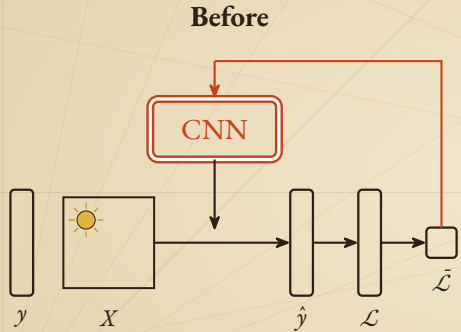
✳ *The grasshoppers, click by click*



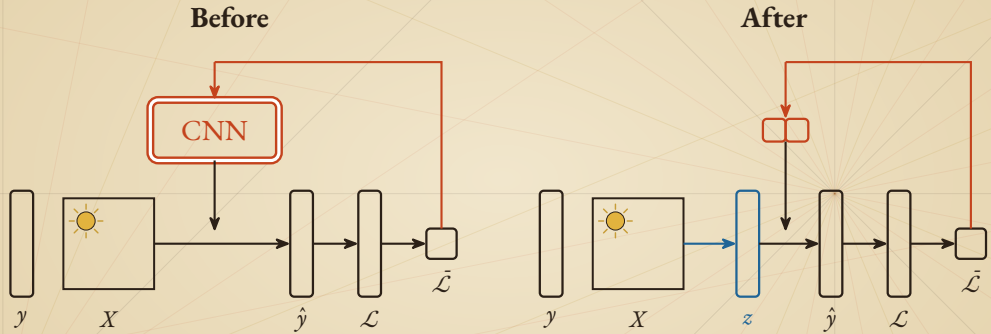
Type where the two grasshoppers start,
then follow them, one click at a time.

https://eth-bmai-hs26.github.io/BMAI-PAGE/viz/we4/grasshopper-regression/

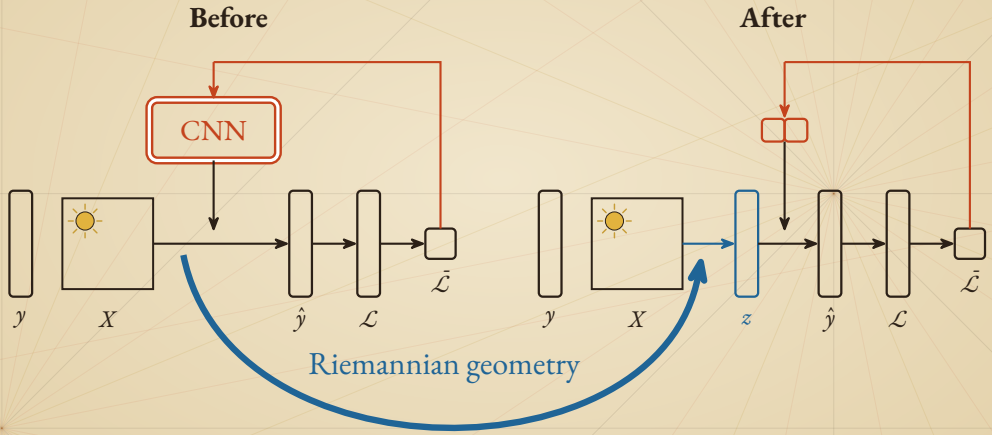
✳ *Before and after*



✳ *Before and after*



✳ *Before and after*





SECTION 2

Charts

- *From the photo to its coordinates, and back*



- *From the photo to its coordinates, and back*



From the photo to its coordinates, and back



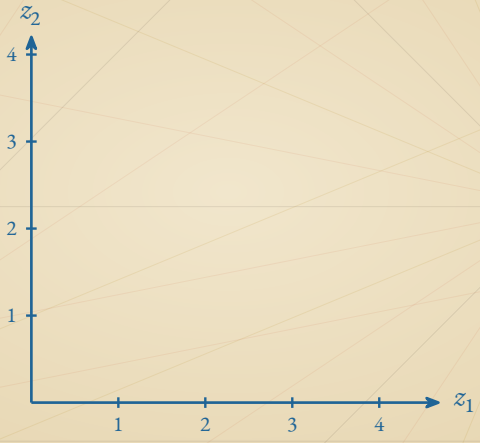
- *From the photo to its coordinates, and back*



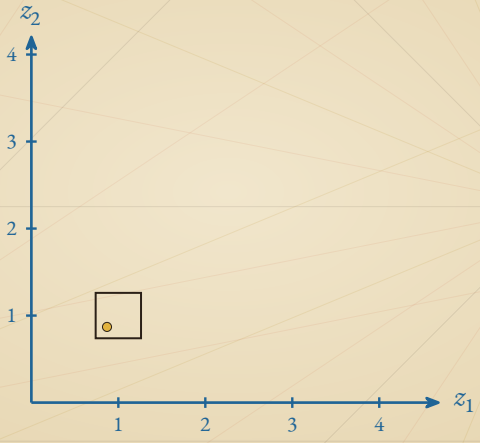
From the photo to its coordinates, and back



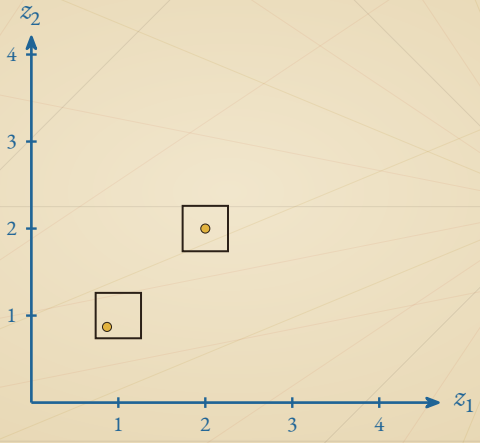
- *The photos, placed at their coordinates*



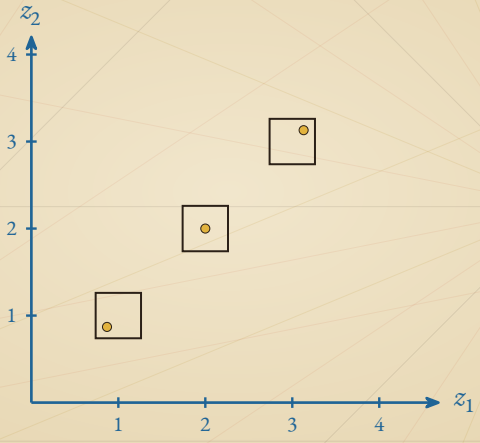
- *The photos, placed at their coordinates*



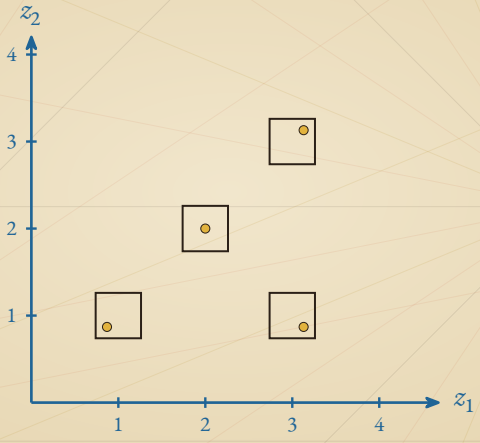
- *The photos, placed at their coordinates*



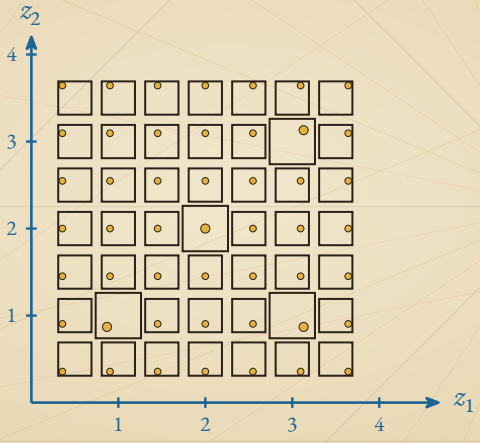
- *The photos, placed at their coordinates*



- *The photos, placed at their coordinates*

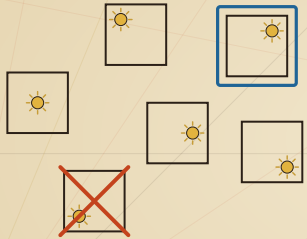


- *The photos, placed at their coordinates*

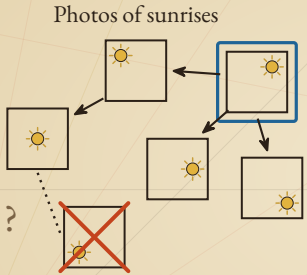


✳ *Walking through the sunrises*

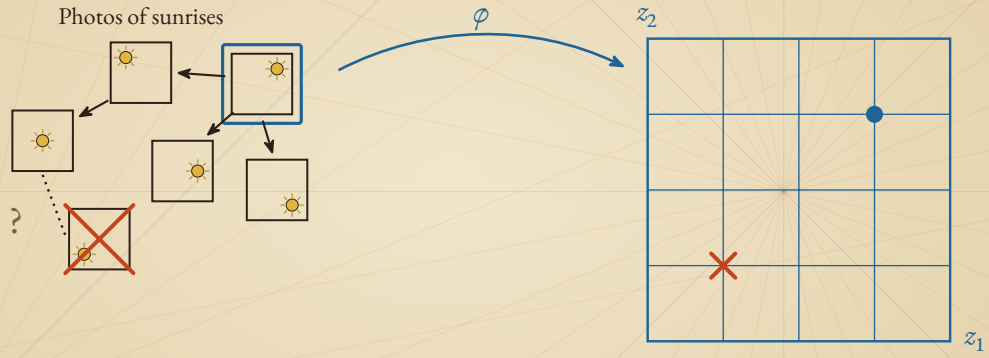
Photos of sunrises



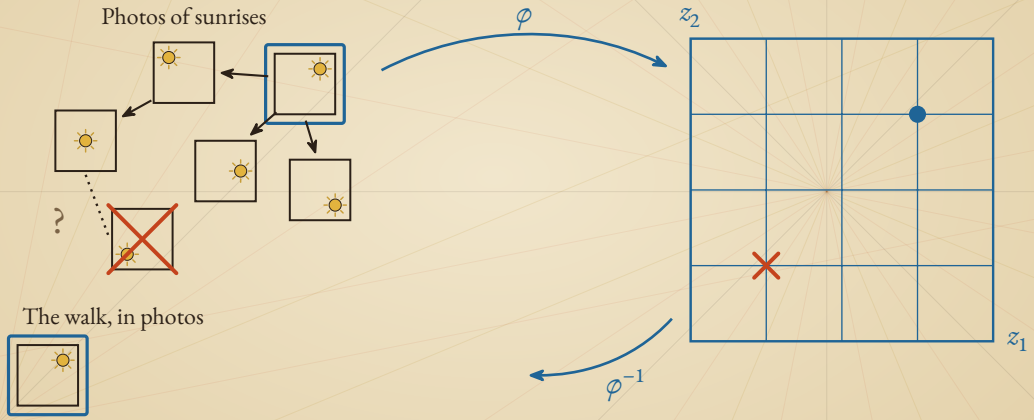
✳ *Walking through the sunrises*



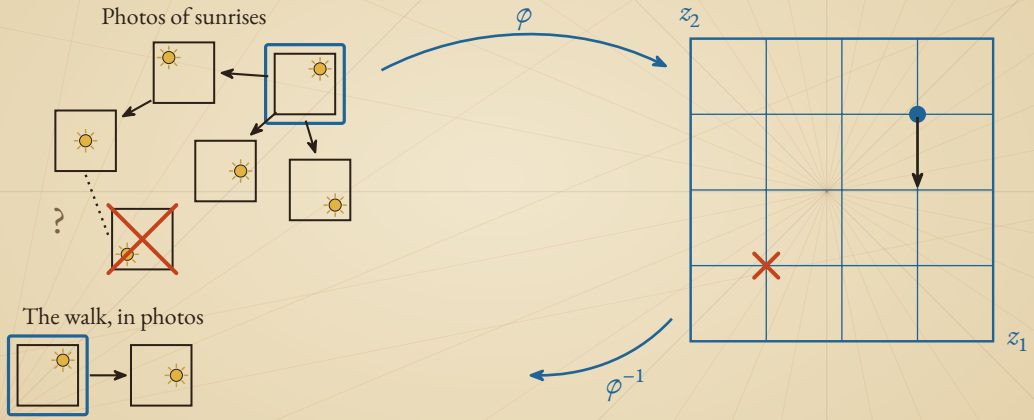
✳ *Walking through the sunrises*



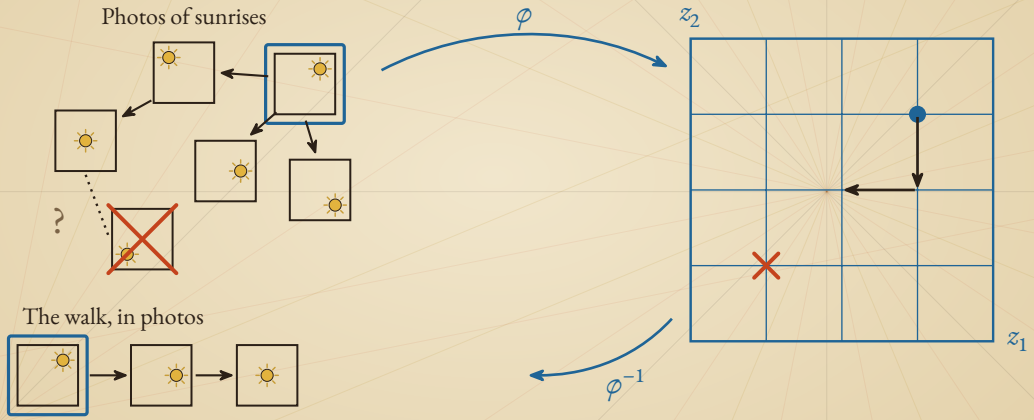
✳ *Walking through the sunrises*



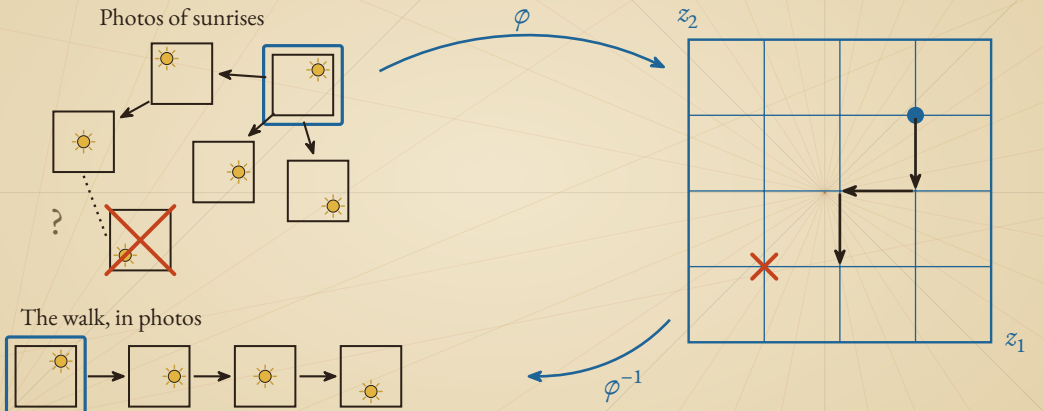
✳ *Walking through the sunrises*



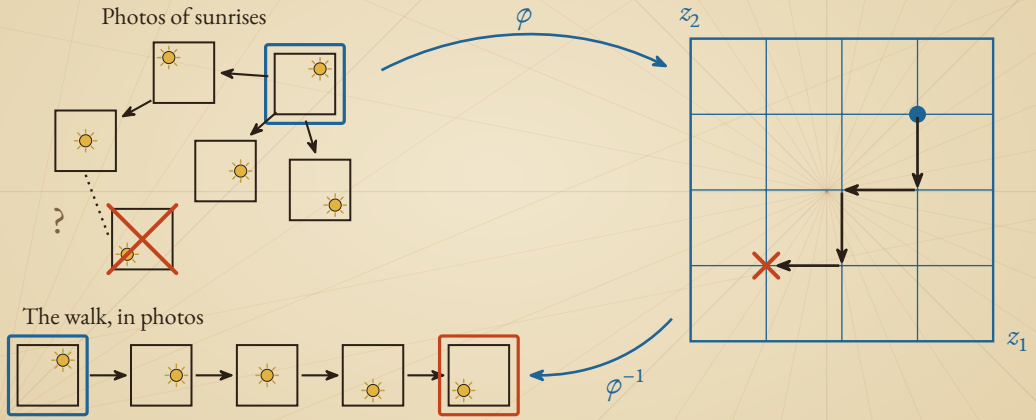
✳ *Walking through the sunrises*



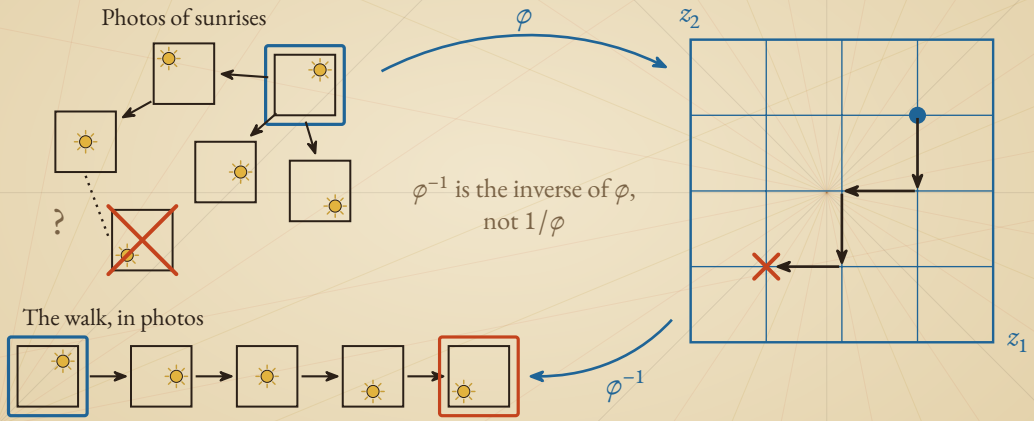
✳ *Walking through the sunrises*



✳ *Walking through the sunrises*



✳ *Walking through the sunrises*





The set of all sunrises forms a *manifold*.

✳ *Manifolds*



The set of all sunrises forms a *manifold*.

A *manifold* is any set of objects that admits a *homeomorphism* to a *navigation chart*.*

* *A white lie: in general, a manifold only needs such a chart around each of its points.*

✳ *Another example*



- *Another example*





✳ *Another example*







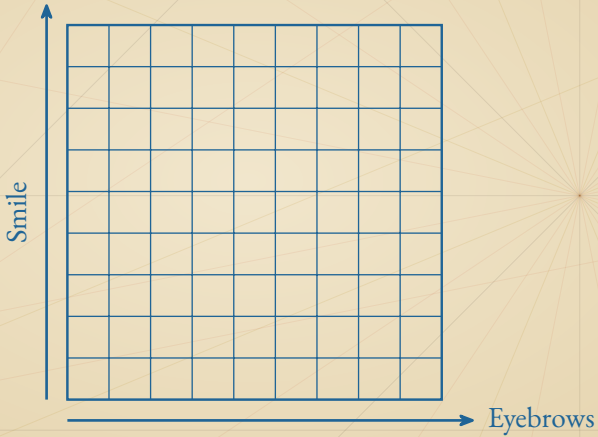
✳ *Another example*



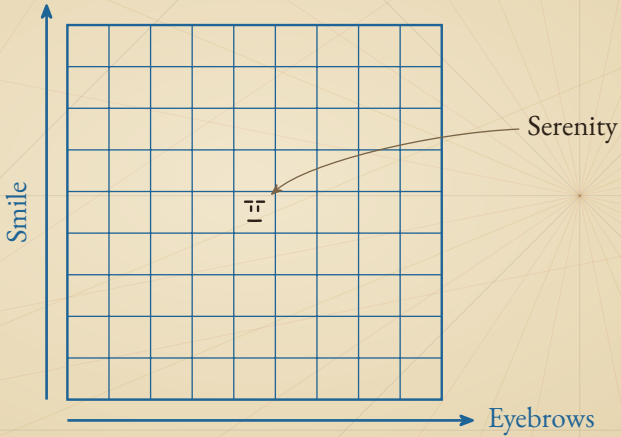
✳ *Another example*



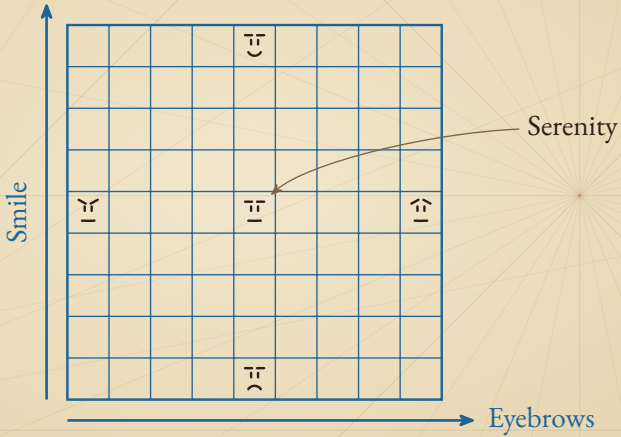
✳ *A chart of faces*



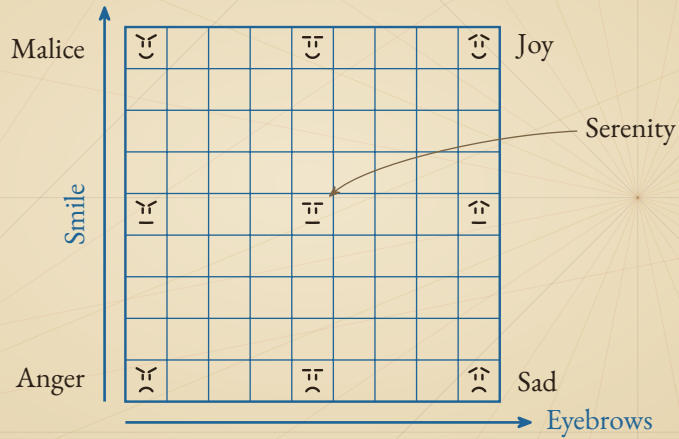
✳ *A chart of faces*



✳ *A chart of faces*



✳ *A chart of faces*



✳ *A chart of faces*

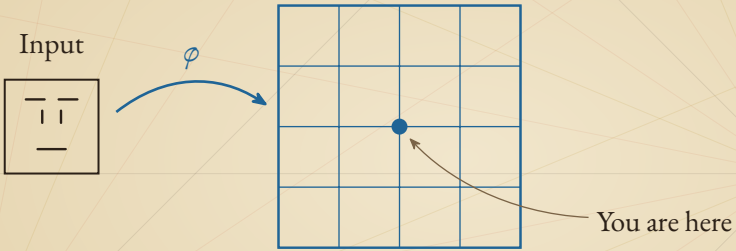


- *The main application of Riemannian geometry to AI*

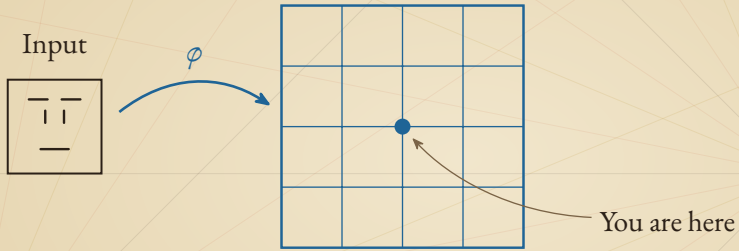
Input



✳ *The main application of Riemannian geometry to AI*



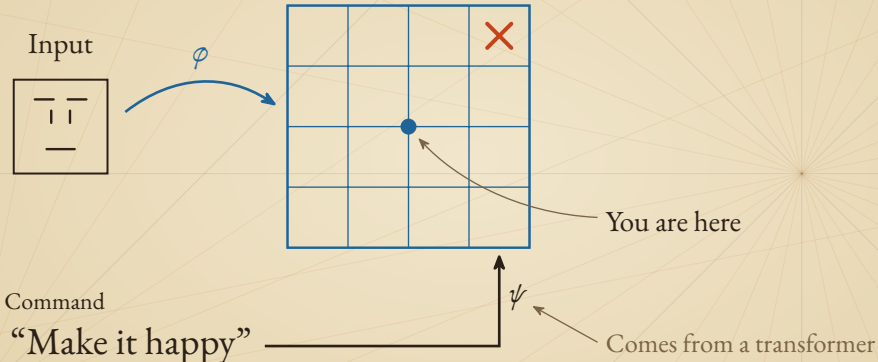
✳ *The main application of Riemannian geometry to AI*



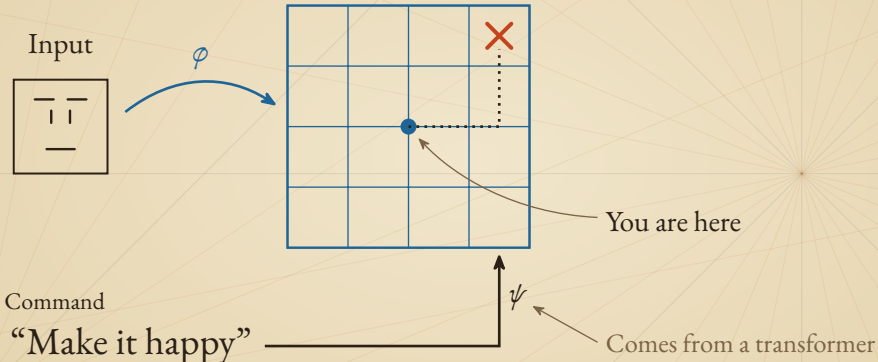
Command

"Make it happy"

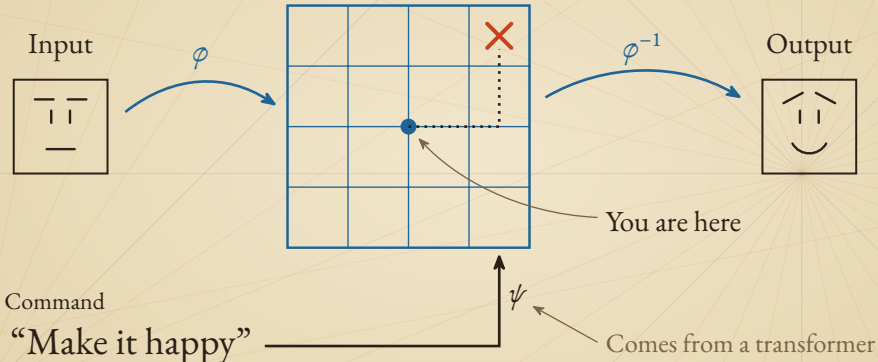
✳ *The main application of Riemannian geometry to AI*



✳ *The main application of Riemannian geometry to AI*



✳ *The main application of Riemannian geometry to AI*

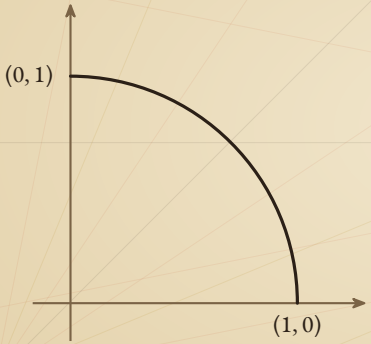




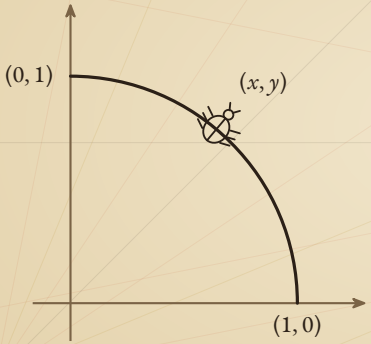
SECTION 3

Examples of manifolds

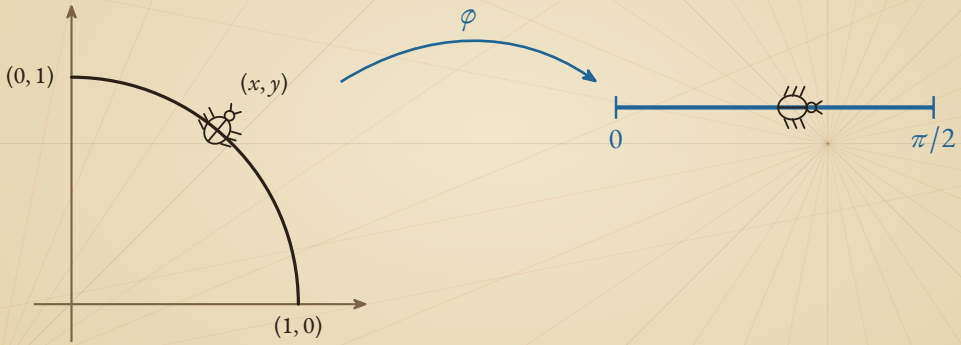
- *An arc of the circle*



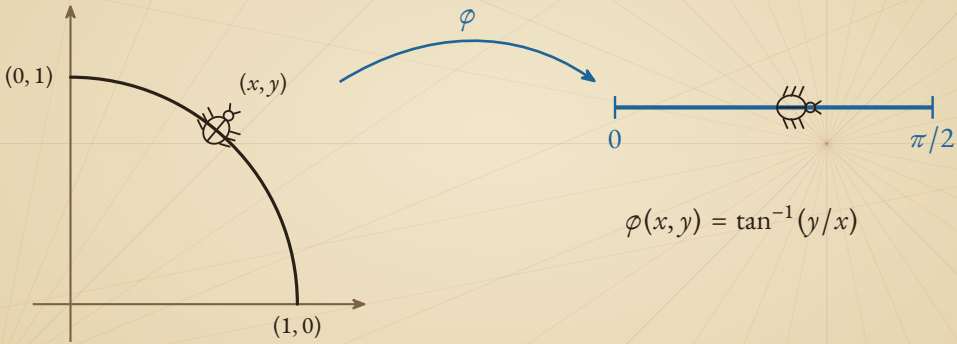
An arc of the circle



✳ *An arc of the circle*

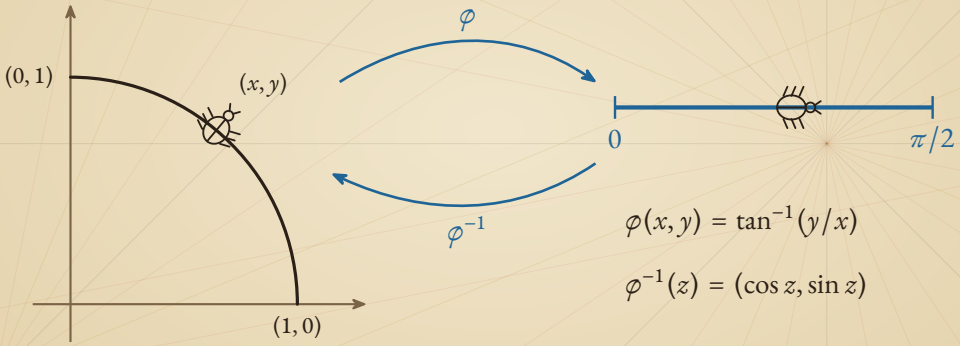


✳ *An arc of the circle*



$$\varphi(x, y) = \tan^{-1}(y/x)$$

✳ *An arc of the circle*

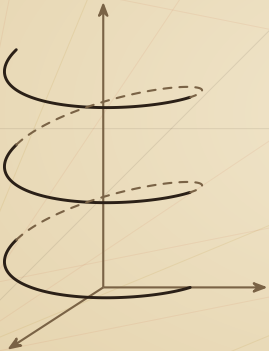


$$\varphi(x, y) = \tan^{-1}(y/x)$$

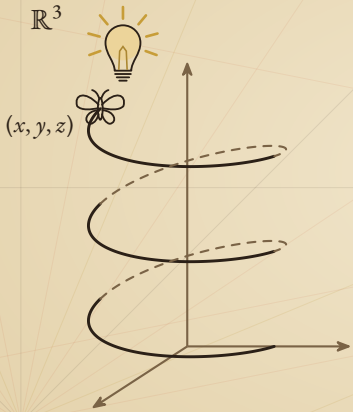
$$\varphi^{-1}(z) = (\cos z, \sin z)$$

✵ *A helix*

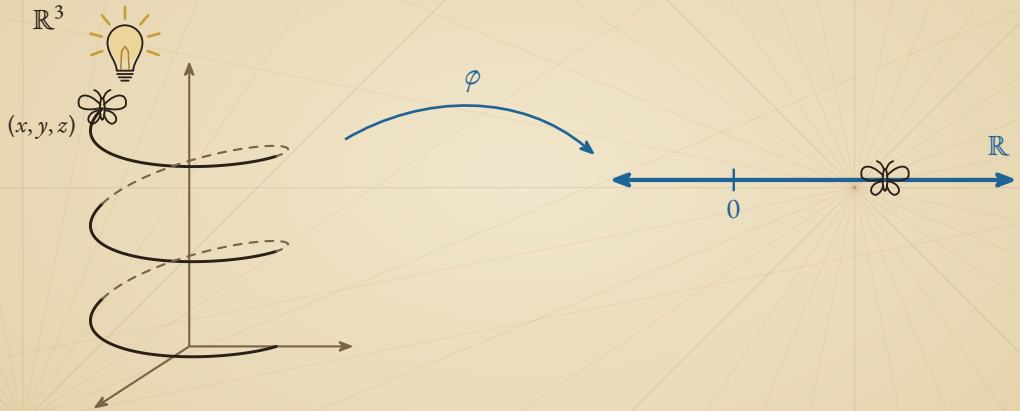
$\mathbb{R}^3$



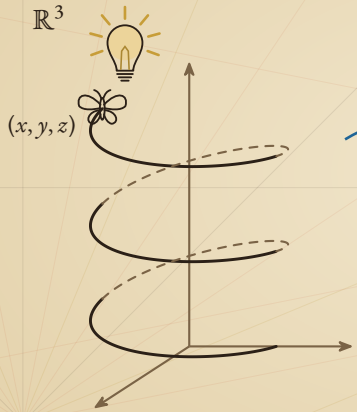
✳ *A helix*



✳ *A helix*



✳ *A helix*



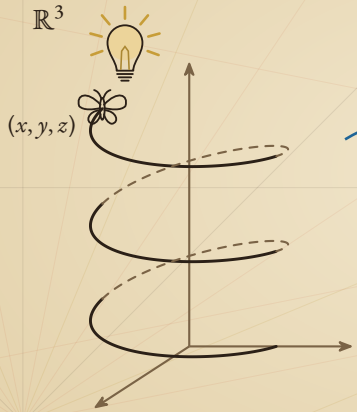
φ



$\varphi^{-1}(t) =$

$\varphi(x, y, z) =$

✳ *A helix*



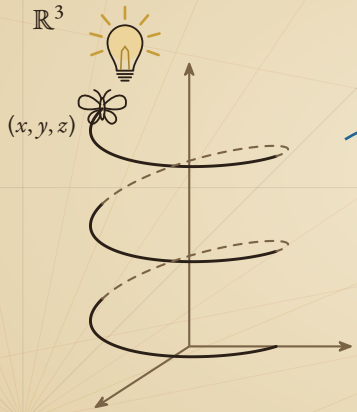
φ

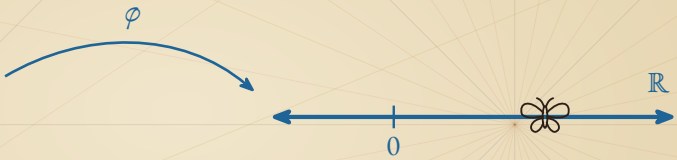


$\varphi^{-1}(t) = (\cos t,\ \sin t,\ t)$

$\varphi(x, y, z) =$

✳ *A helix*





$$\varphi^{-1}(t) = (\cos t,\ \sin t,\ t)$$

$$\varphi(x, y, z) = z$$

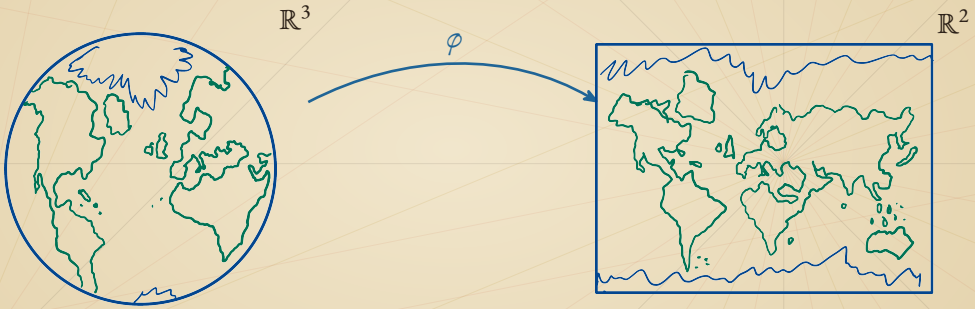
✳ *The globe and the world map*

$\mathbb{R}^3$

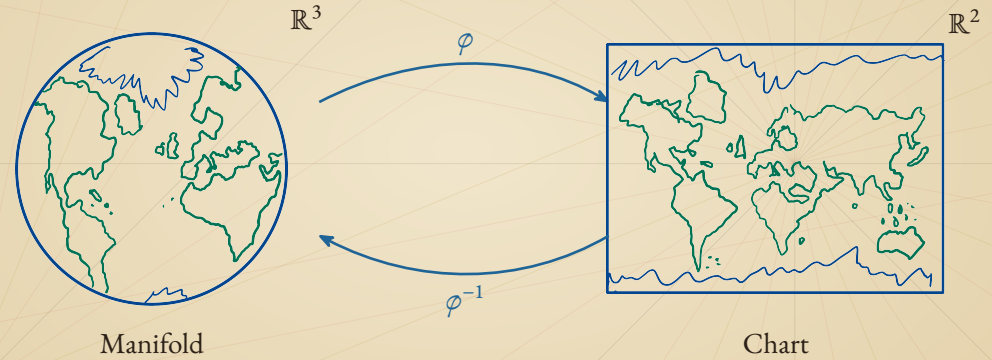


Manifold

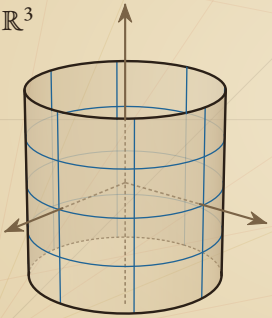
✳ *The globe and the world map*



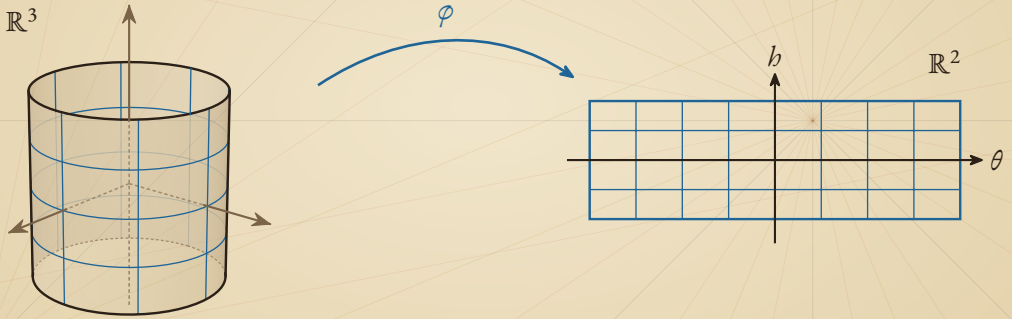
✳ *The globe and the world map*



✳ *A cylinder*

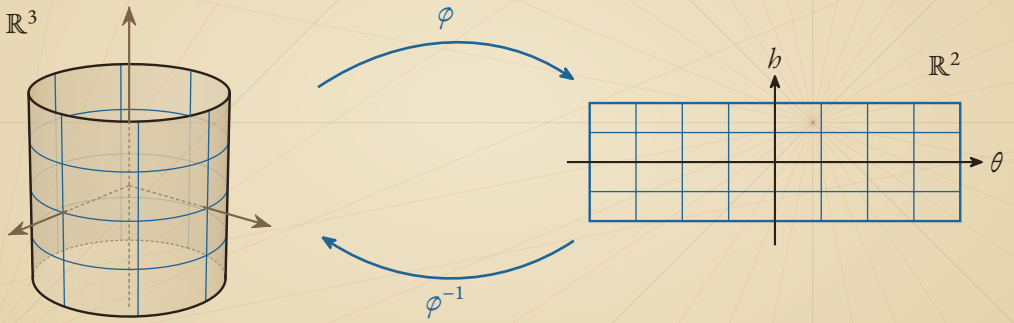


✳ *A cylinder*

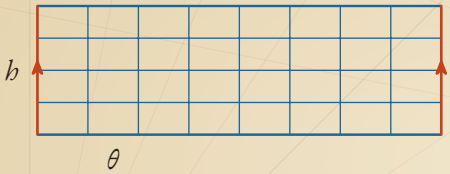


✳ *A cylinder*

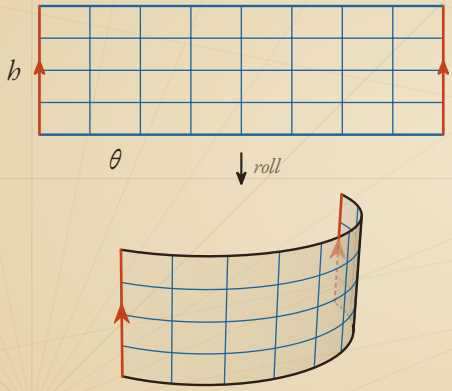
$$\varphi^{-1}(\theta, h) = \left(\cos\tfrac{2\pi\theta}{360},\ \sin\tfrac{2\pi\theta}{360},\ h\right)$$



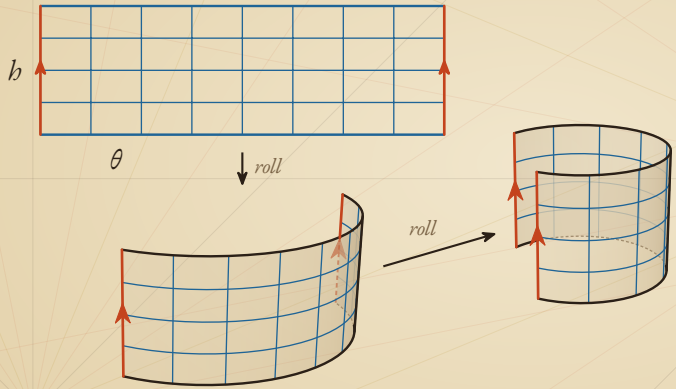
✳ *Folding a chart into a cylinder*



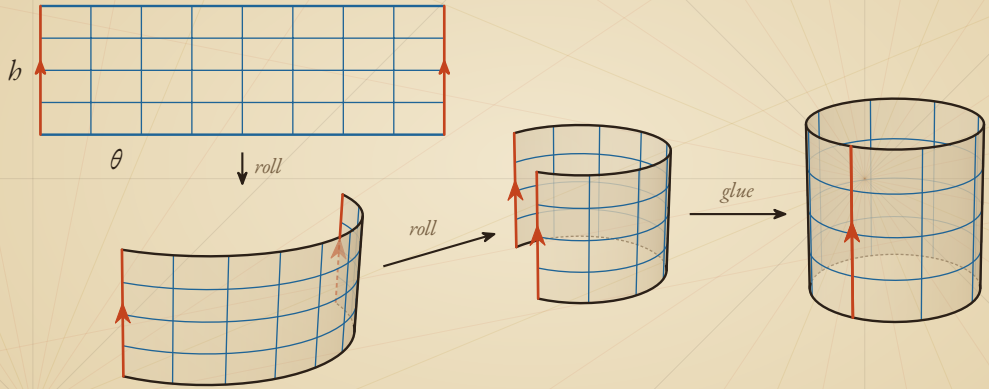
✳ *Folding a chart into a cylinder*



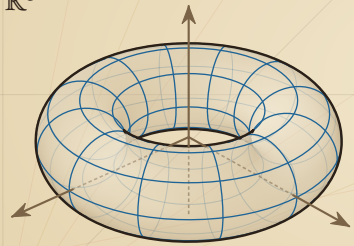
✳ *Folding a chart into a cylinder*



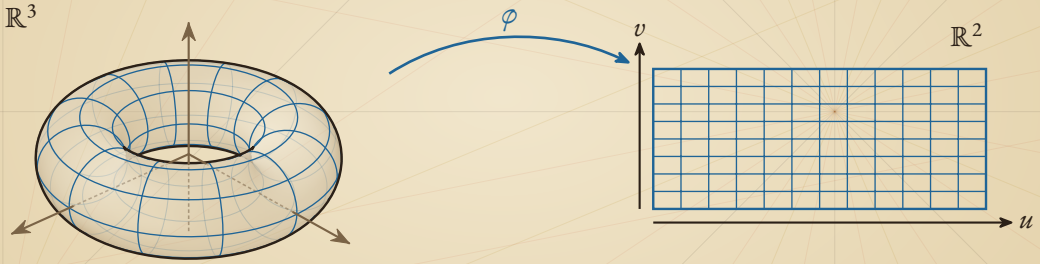
✳ *Folding a chart into a cylinder*



✳ *A torus*



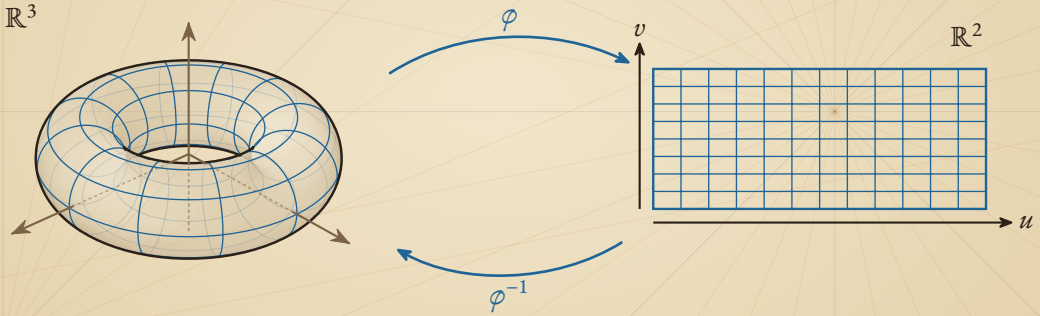
✳ *A torus*



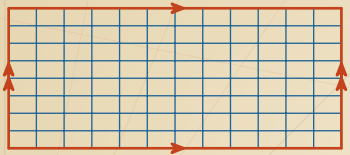
✳ *A torus*

$$\varphi^{-1}(u, v) = \big((R + r\cos v)\cos u,\ (R + r\cos v)\sin u,\ r\sin v\big)$$

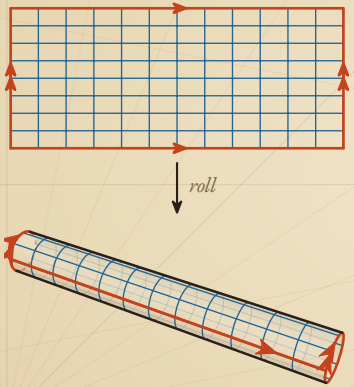
R: centre to the middle of the tube. r: radius of the tube.



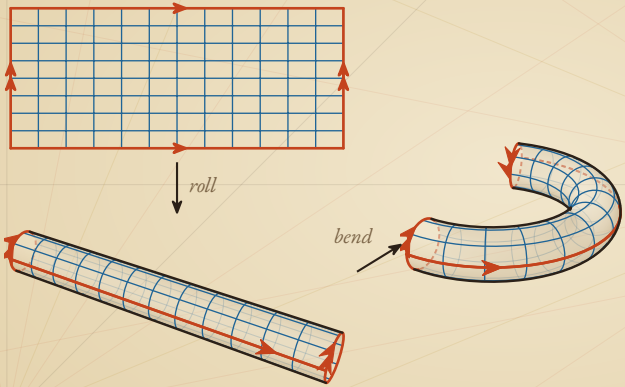
✳ *Folding a chart into a torus*



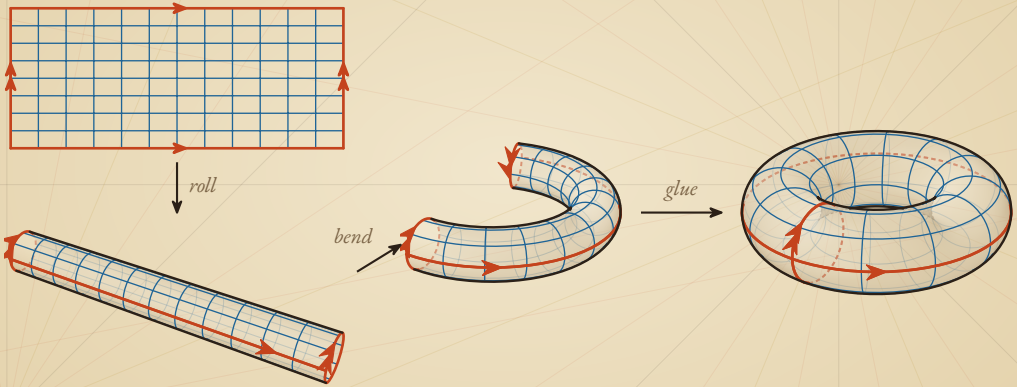
✳ *Folding a chart into a torus*



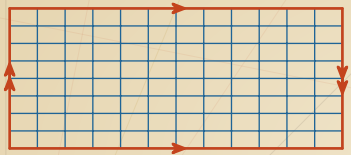
✳ *Folding a chart into a torus*



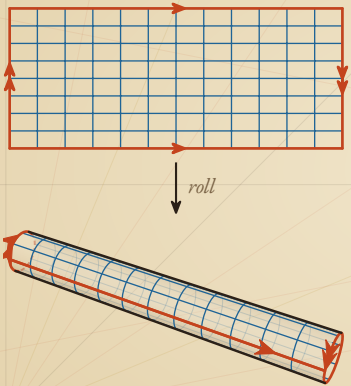
✳ *Folding a chart into a torus*



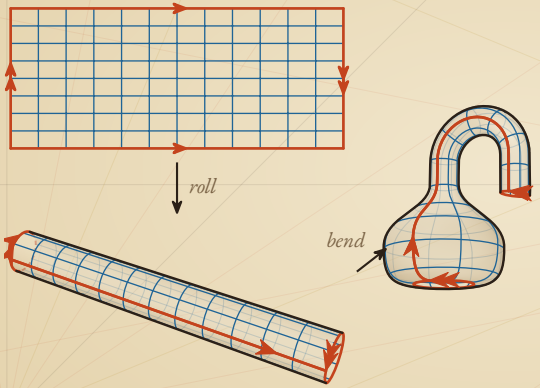
✳ *Folding a chart into a Klein bottle*



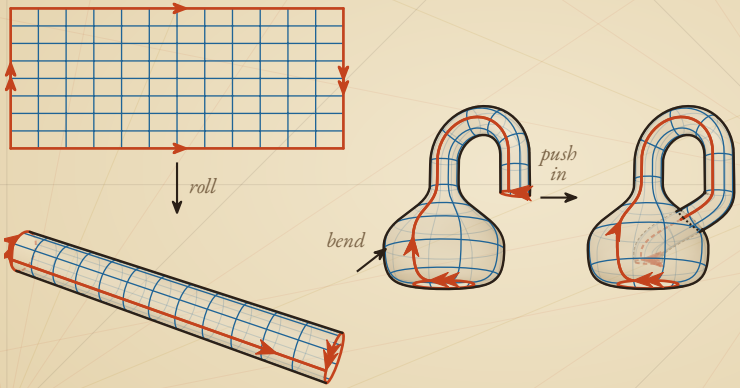
✳ *Folding a chart into a Klein bottle*



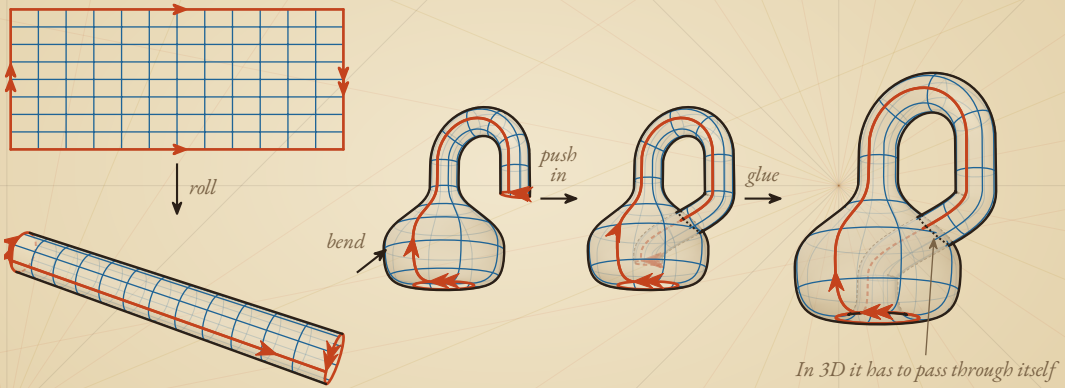
✳ *Folding a chart into a Klein bottle*



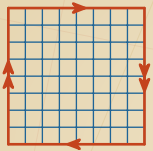
✳ *Folding a chart into a Klein bottle*



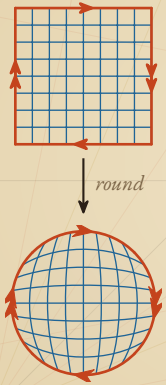
✳ *Folding a chart into a Klein bottle*



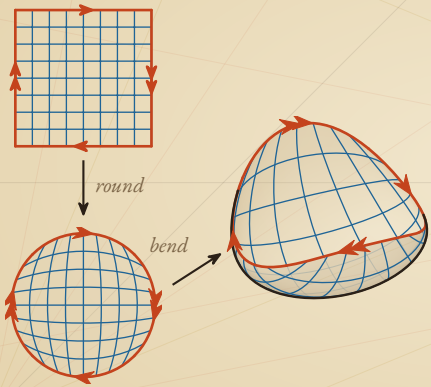
- *Folding a chart into a projective plane*



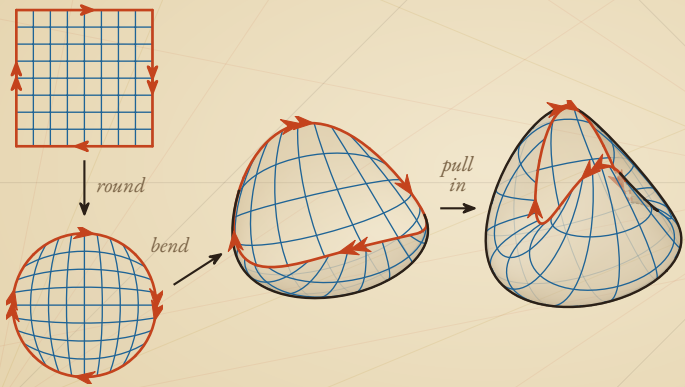
✳ *Folding a chart into a projective plane*



✳ *Folding a chart into a projective plane*



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